MLOps CI/CD Pipeline Project Report

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Objective

The objective of this project is to implement a robust, production-ready continuous integration and continuous delivery (CI/CD) pipeline for a machine learning workflow. This pipeline automates testing, code validation, build creation, and packaging stages to guarantee seamless code quality and artifact maintenance on every single commit.

MLOps CI/CD Pipeline Architecture

The automation pipeline is built using GitHub Actions and triggered on pushes and pull requests to the main branch. The workflow is split into two distinct stages:

1. Continuous Integration (CI): Clones the code, sets up the Python environment, installs specified library dependencies via pip, and executes unit tests using pytest to verify core algorithm stability.

2. Continuous Delivery (CD): Once tests pass, it packages the source scripts, configuration configurations, and dependency matrices into a structured zip file artifact and uploads it to GitHub Actions Storage.

Experiment and Unit Test Results

A Logistic Regression classifier was trained on the standardized Iris flowers classification dataset. The unit test suite evaluates the model's prediction capabilities against a designated threshold limit (90% target classification accuracy score).

Evaluation Result: All tests passed successfully with an evaluated model accuracy score exceeding 95% on the test subset.

Conclusion

Integrating automated validation protocols prevents regression defects from impacting production deployments. By isolating environments and enforcing testing thresholds dynamically on cloud runners, we ensure the integrity and predictability of machine learning applications in continuous-delivery systems.